University of Florida, Department of Mathematics

Logic PhD Exam, May 2017

Answer six questions, at least one in each section.

Section 1

1. State and prove Tarski's criterion for elementarity of submodels.

2. State and prove the downward Löwenheim–Skolem theorem.

3. Prove that the models $\langle \mathbb{Z}, \leq \rangle$ (integers with the usual ordering) and $\langle \mathbb{Z} + \mathbb{Z}, \leq \rangle$ (two copies of integers with the usual ordering, one following the other) are elementarily equivalent.

Section 2

1. Prove that for any cardinal κ, the cofinality $\mathrm{cf}(2^\kappa) > \kappa$.

2. Show that the wellordering principle implies the Zorn's lemma.

3. Show that a c.c.c. forcing preserves all cardinals.

Section 3

1. Show that every infinite computably enumerable set has an infinite computable subset.

2. Show that there are two subsets of ω which are incomparable in the sense of the Turing ordering.

3. Explain what a Π^1_1 complete set is and provide an example with a proof.